SPA617 - RADIO ASTRONOMY LABORATORY

Experiment: RF Transmission, Reflections and Standing Waves

1. Aim

To understand signal propagation, reflection and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2. Objectives

1. Determine the phase velocity and velocity factor from the frequency spacing between consecutive maxima or minima.
2. Study reflection from an open-ended cable and a 50 ohm terminated cable.
3. Estimate the reflection coefficient and verify approximate matching to 50 ohm.
4. Calculate inductance and capacitance per unit length.
5. Measure insertion loss and attenuation per unit length.

3. Apparatus

RF signal generator, digital storage oscilloscope, spectrum analyser, RF Tee connector, coaxial cable under test, short reference cable, 50 ohm termination and suitable RF adapters/connectors.

4. Theory and Formulae

For a reflected signal, the amplitude at the Tee junction varies periodically with frequency because of interference between incident and reflected waves.

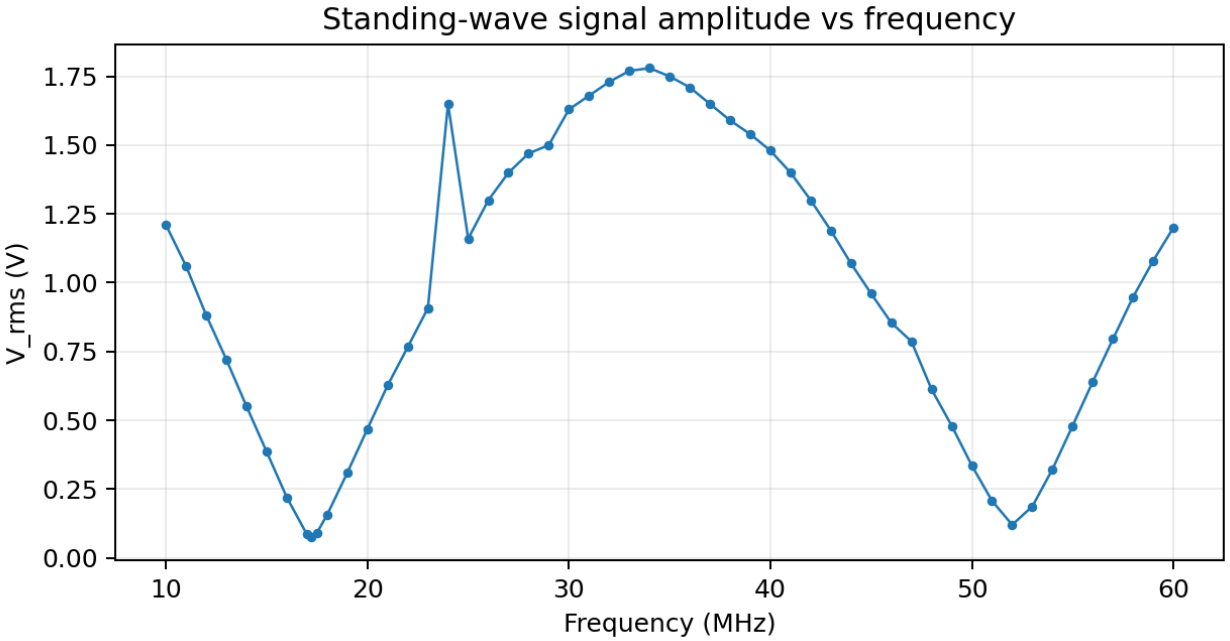
$\Delta f = |f_2 - f_1|$
$v_p = 2\ell\Delta f$
$VF = v_p / c$
$|\Gamma| = (Vmax - Vmin) / (Vmax + Vmin)$
$L' = Z_0 / v_p$
$C' = 1 / (Z_0 v_p)$
$IL = P_{ref} - P_{cable}$
$\alpha = IL / \ell$

5. Part A - Standing-Wave Measurement

Cable length: ℓ = 305 cm = 3.05 m.

The following readings were recorded during the frequency sweep.

Part A: Standing-Wave Signal Amplitude vs Frequency



Frequency (MHz)	V_rms (V)
10	1.21
11	1.06
12	0.8816
13	0.72045
14	0.55041
15	0.385
16	0.21936
17	0.0847
17.2	0.07608
17.5	0.0919
18	0.15607
19	0.30907
20	0.46867
21	0.62676
22	0.76622
23	0.90745
24	1.65
25	1.16
26	1.3
27	1.4
28	1.47
29	1.5
30	1.63
31	1.68
32	1.73
33	1.77
34	1.78
35	1.75
36	1.71

Frequency (MHz)	V_rms (V)
37	1.65
38	1.59
39	1.54
40	1.48
41	1.4
42	1.3
43	1.19
44	1.07
45	0.9601
46	0.8545
47	0.7866
48	0.612
49	0.48
50	0.335
51	0.207
52	0.12
53	0.187
54	0.322
55	0.48
56	0.64
57	0.795
58	0.948
59	1.08
60	1.2

6. Part B - Reflection Coefficient

Measured extrema from the notebook: open circuit Vmax = 1.72 V, Vmin = 0.0953 V; 50 ohm load Vmax = 1.06 V, Vmin = 0.93336 V.

Termination	Vmax (V)	Vmin (V)	|Γ|
Open circuit	1.72	0.0953	0.8950
50 ohm load	1.06	0.93336	0.0635

Open circuit: $|\Gamma_{open}| = (1.72 - 0.0953)/(1.72 + 0.0953) = 0.8950$.

50 ohm load: $|\Gamma_{50}| = (1.06 - 0.93336)/(1.06 + 0.93336) = 0.0635$.

The reflection is greatly reduced with the 50 ohm termination, indicating that the cable is approximately matched to 50 ohm.

7. Part C - Inductance and Capacitance per Unit Length

Using the recorded phase velocity $v_p = 2.10 \times 10^8$ m/s and $Z_0 \approx 50$ ohm:

$L' = 50/(2.10 \times 10^8) = 2.381\text{e-}07$ H/m = 238.1 nH/m.

$C' = 1/[50 \times (2.10 \times 10^8)] = 9.524\text{e-}11$ F/m = 95.24 pF/m.

8. Part D - Cable Attenuation

The notebook records the Part-D cable length as approximately 91.5 cm = 0.915 m.

Freq (MHz)	P_ref (dBm)	P_cable (dBm)	IL (dB)	α (dB/m)
100	-1.40	-1.65	0.25	0.273
150	-1.40	-1.80	0.40	0.437
200	-1.40	-1.96	0.56	0.612
250	-1.40	-2.12	0.72	0.787
300	-1.40	-2.18	0.78	0.852

9. Calculations and Final Results

From the marked minima in the Part-A readings, $f_1 = 17.2$ MHz and $f_2 = 52.0$ MHz.

$\Delta f = 52.0 - 17.2 = 34.8$ MHz.

$v_p = 2 \times 3.05 \times 34.8 \times 10^6 = 2.12 \times 10^8$ m/s.

$VF = (2.12 \times 10^8)/(3.00 \times 10^8) = 0.708 \approx 0.71$.

The notebook separately records $v_p \approx 2.10 \times 10^8$ m/s and VF ≈ 0.66. The direct calculation above is based on the marked minima and the stated 305 cm cable length.

Quantity	Final value
Cable length (Part A)	3.05 m
Frequency spacing, Δf	34.8 MHz
Phase velocity, v_p	2.12×10^8 m/s
Velocity factor, VF	0.708 ≈ 0.71
Reflection coefficient, open	0.8950
Reflection coefficient, 50 ohm load	0.0635
Approx. characteristic impedance	≈ 50 ohm
Inductance per unit length, L′	238.1 nH/m
Capacitance per unit length, C′	95.24 pF/m
Part-D cable length	0.915 m

10. Conclusion

The frequency sweep shows periodic variation in signal amplitude due to interference between incident and reflected waves. The marked minima give a frequency spacing of 34.8 MHz, corresponding to a phase velocity of about 2.12×10^8 m/s. The reflection coefficient is strongly reduced when a 50 ohm termination is connected, supporting approximate matching to a 50 ohm transmission line. The calculated inductance and capacitance per unit length are 238.1 nH/m and 95.24 pF/m, respectively. The attenuation measurements show increasing cable loss with frequency.

11. Precautions / Sources of Uncertainty

• Keep the signal-generator amplitude constant during the frequency sweep.
• Use 1 Mohm oscilloscope input so that the Tee junction is not significantly loaded.
• Use the same adapters and connector arrangement for reference and cable measurements.
• Avoid overdriving the oscilloscope or spectrum analyser.
• Reading uncertainty, connector/adaptor losses, cable-length uncertainty and instrument calibration may affect the results.

12. Data Note

The report uses the readings visible in the uploaded notebook. The notebook gives 305 cm for Part A and approximately 91.5 cm for Part D; these are kept separate. It also contains a separately written approximate phase velocity of 2.10×10^8 m/s. The numerical calculations are shown explicitly so that the result can be checked from the recorded readings.